ROOM LIGHT CONTROL SYSTEM

id091800012300

Outline

- The room light control system changes the illumination condition and illumination level of the interior lights (door position) according to whether the doors are opened/closed and locked/unlocked.

Function

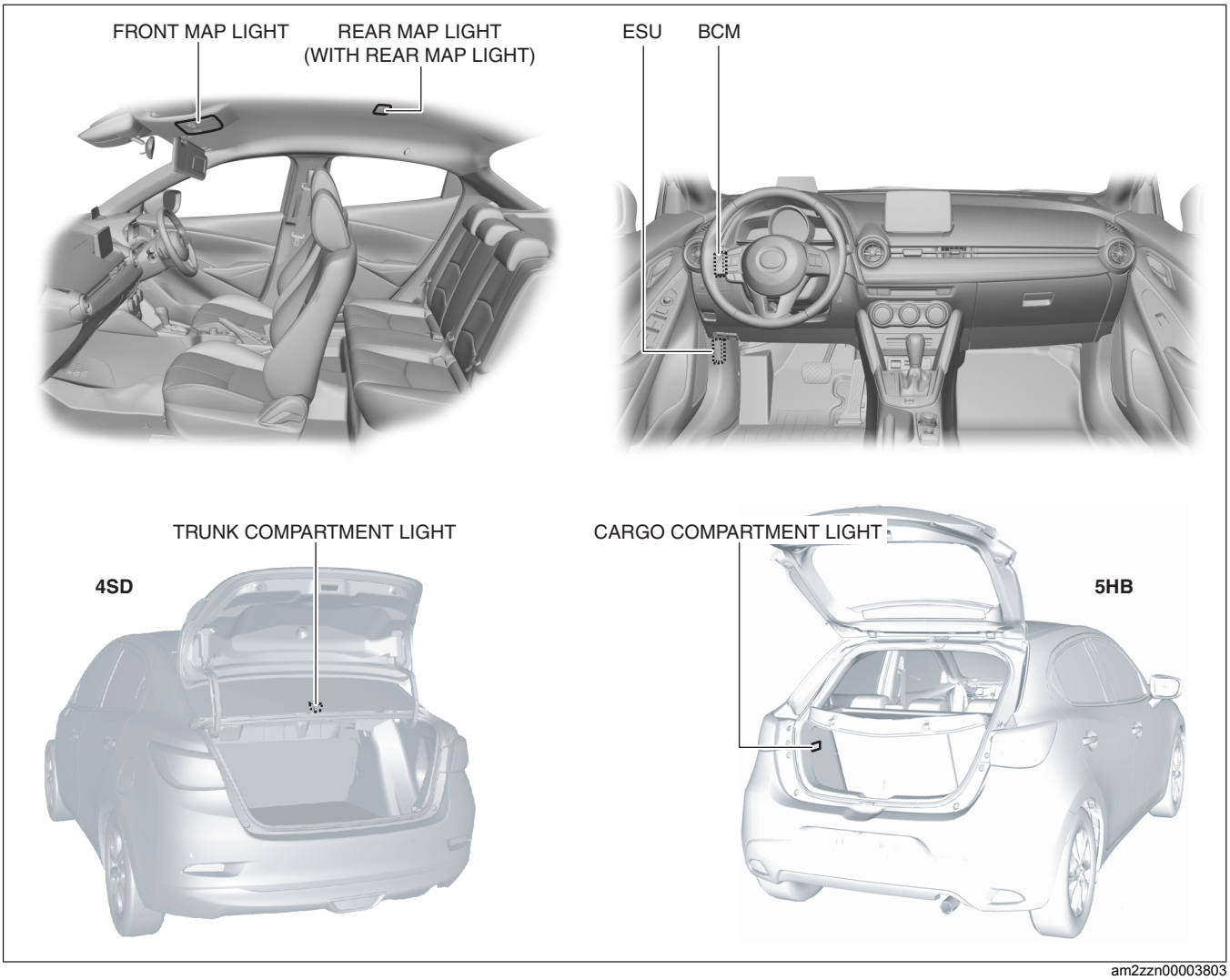
- The BCM changes the illumination condition of the map light and trunk compartment light (4SD)/cargo compartment light (5HB) and illumination level of the map light based on the following signals:

Signal name	Sending module/part name	Communication method
Door latch switch signal	Door latch switch	Hard wire
Door lock-link switch signal	Front door lock-link switch (driver-side)	
Trunk lid latch switch signal (4SD)	Trunk lid latch switch	
Liftgate latch switch signal (5HB)	Liftgate latch switch	
Ignition switch position signal	IG1 relay	
	Instrument cluster	HS-CAN

- The electrical supply unit (ESU) turns the map light and trunk compartment light (4SD)/cargo compartment light (5HB) on/off based on the following signals:

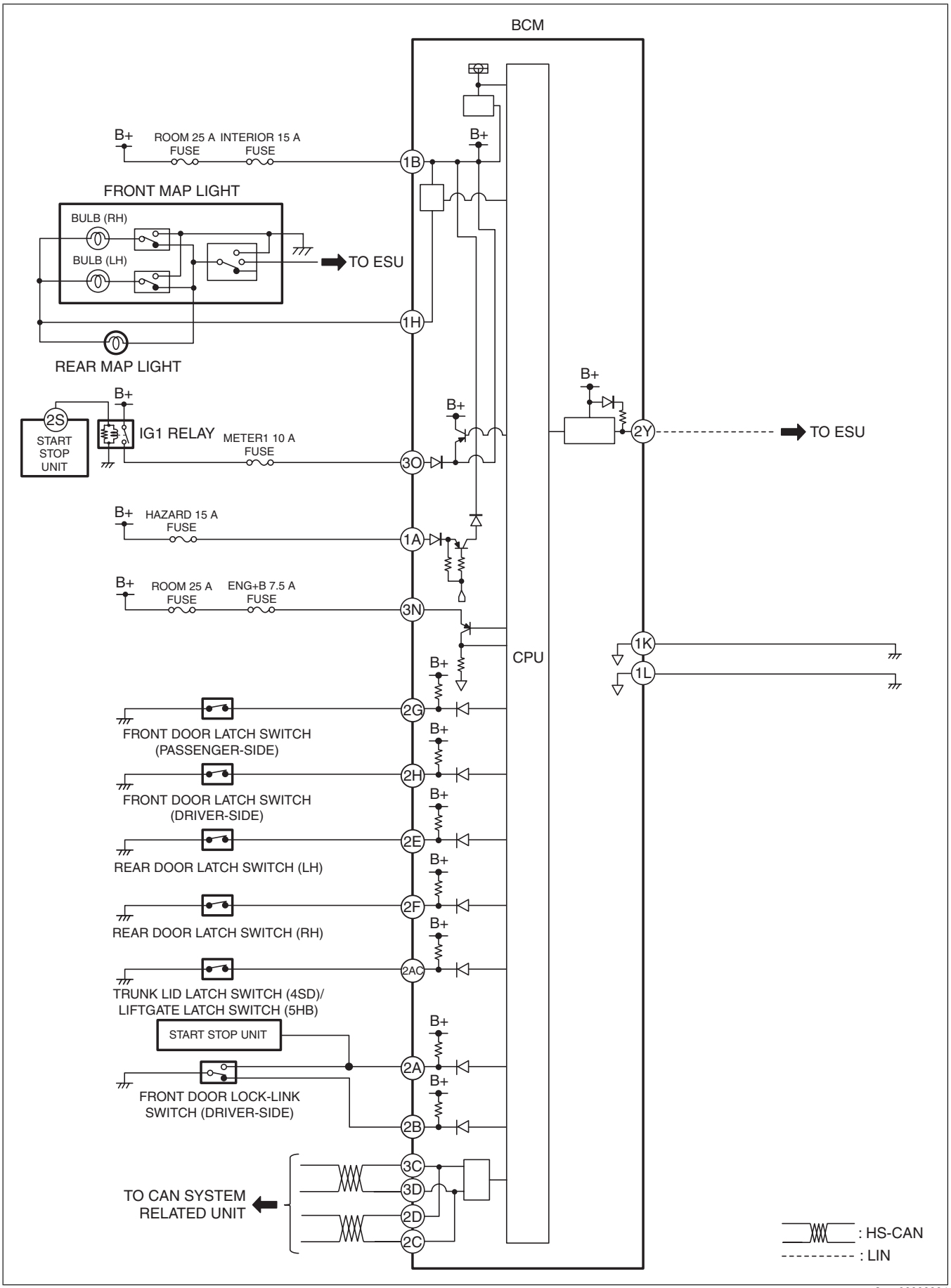
Signal name	Sending module/part name	Communication method
Map light on request signal	BCM	LIN
Trunk compartment light on request signal (4SD)		
Cargo compartment light on request signal (5HB)		

Structural View

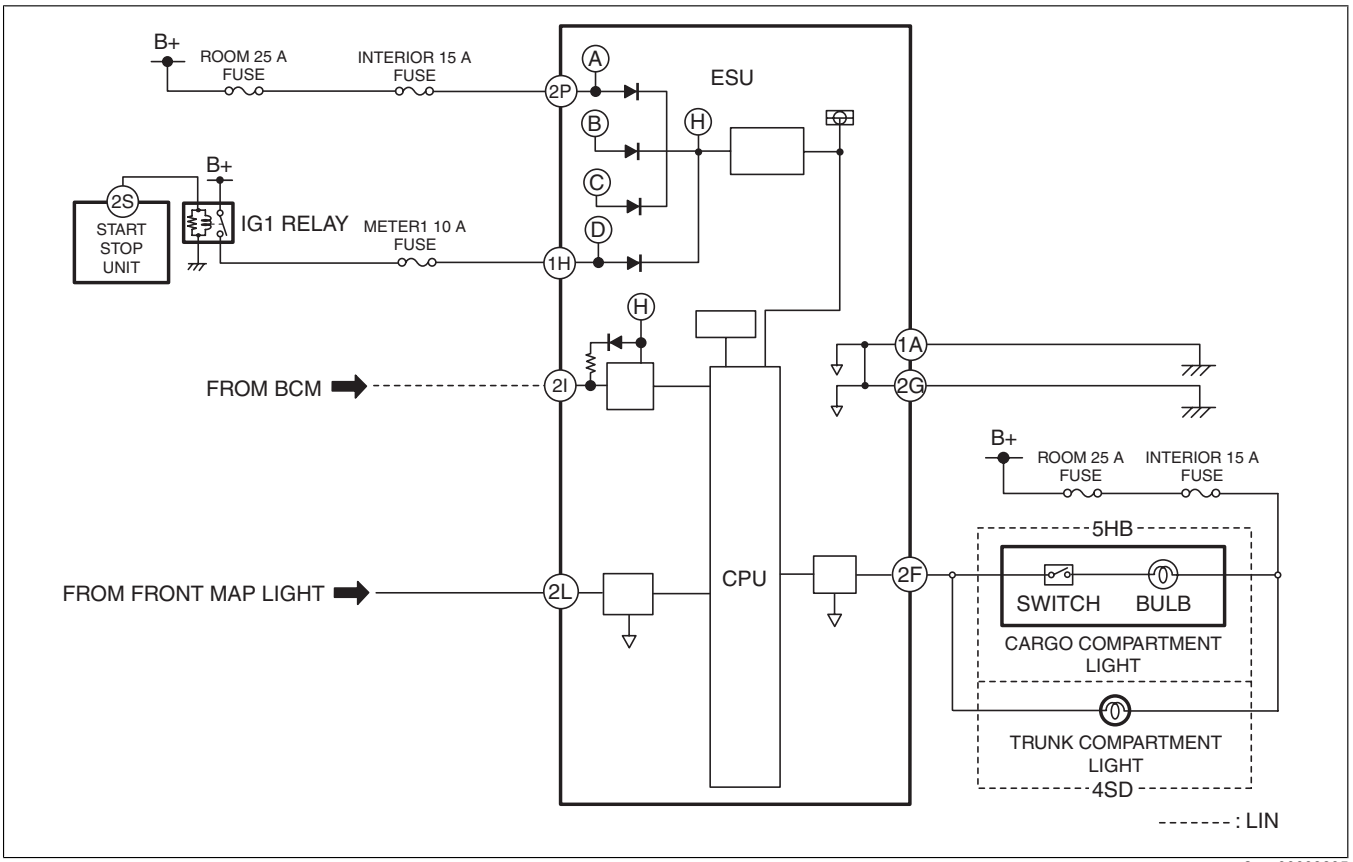


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System Wiring Diagram



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Operation

Operation Conditions	Interior light		Lights off condition (Any of the following conditions is met)
	Illumination period	Brightness level	
With ignition switched off or ACC and all doors closed (all door latch switches are off), a condition is detected when any door is opened (any door latch switch is on)	**Approx. 30 min** *1	100 %	• After illumination time elapses*2 • All doors other than trunk lid (4SD)/liftgate (5HB) closed (all door latch switches off)
With any door opened (any door latch switch is on) while ignition is switched off or ACC, a condition is detected when all doors are closed (all door latch switches are off)	• **Approx. 5 s** *3 • **Approx. 15 s** *1 *4	60 %	• After illumination time elapses • Driver's door lock knob locked (door lock-link switch at LOCK position) • Ignition switched ON (engine off or on)
With all doors closed (all door latch switches are off) while ignition is switched off or ACC, the driver's door lock knob is switched from lock to unlock	**Approx. 30 s**	100 %	• After illumination time elapses • Driver's door lock knob locked (door lock-link switch at LOCK position) • Ignition switched ON (engine off or on)
With all doors closed (all door latch switches are off) while ignition is switched ACC or ON (engine off or on), the ignition switched off condition is detected	**Approx. 15 s**	100 %	• After illumination time elapses • Driver's door lock knob locked (door lock-link switch at LOCK position) • Ignition switched ON (engine off or on)
• Trunk lid is open (4SD) • Liftgate is open with cargo compartment light switch on (5HB)	**Approx. 30 min** *1	100 %	• After illumination time elapses*5 • Trunk lid (4SD)/liftgate (5HB) closed • Cargo compartment light switch turned off (5HB)

*1 : Illumination time can be changed using the Mazda Modular Diagnostic System (M-MDS) or center display (with center display). (See LIGHTING SYSTEM PERSONALIZATION.)

*2 : When any of the following conditions is met after the interior lights turn off according to the off conditions, the lights turn on again.

— The ignition is switched ON (engine off or on).
— After closing all doors, any door is opened again. (All door latch switches off, then any door latch switch is on)

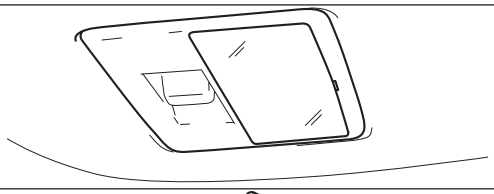
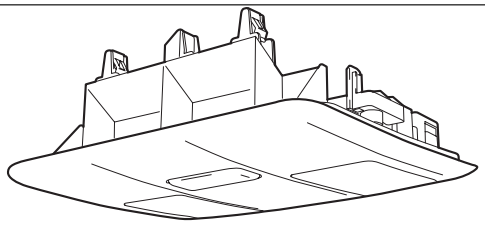
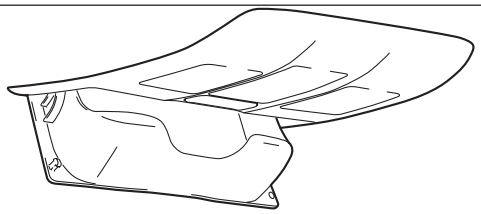
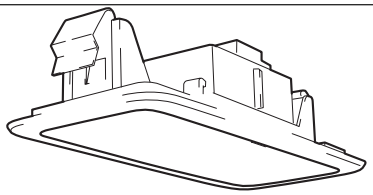
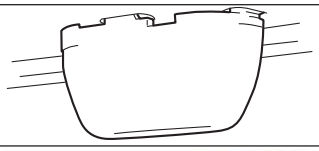
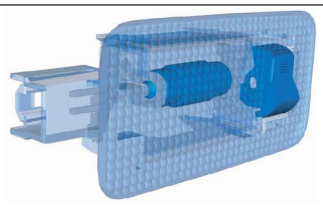
*3 : No remote transmitter inside vehicle

*4 : Remote transmitter inside vehicle

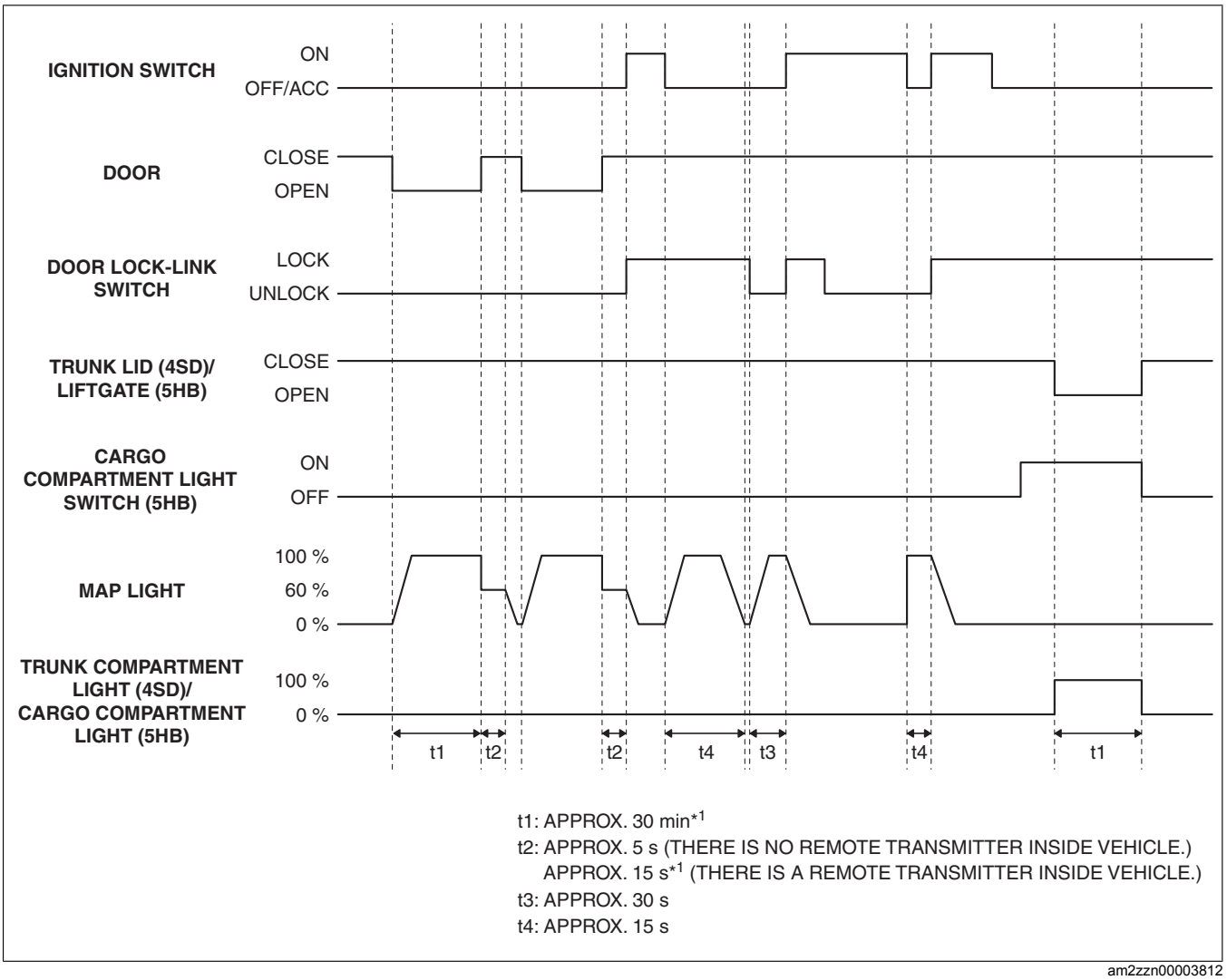
*5 : When any of the following conditions is met after the trunk compartment light (4SD)/cargo compartment light (5HB) turns off according to the off conditions, the light turns on again.

- The ignition is switched ON (engine off or on).
- After trunk lid is closed, and then opened again. (Trunk lid latch switch off, then switch on) (4SD)
- After liftgate is closed, and then opened again. (Liftgate latch switch off, then switch on) (5HB)
- Any door is opened/closed. (Any door latch switch on, then switch off)
- Door is locked/unlocked. (Door lock-link switch locked, then switch unlocked)

×: With room light control system
—: No system

Type			Installation position	Room light control system
Front map light	Single light type		Front area of roof	×
	Dual light type (without sunglasses holder)			
	Dual light type (with sunglasses holder)			
Rear map light (with rear map light)			Center area of roof	×
Trunk compartment light (4SD)			Trunk compartment	×
Cargo compartment light (5HB)			Cargo compartment	×*1

*1 : Cargo compartment light switch is on



t1: APPROX. 30 min*1
t2: APPROX. 5 s (THERE IS NO REMOTE TRANSMITTER INSIDE VEHICLE.)
APPROX. 15 s*1 (THERE IS A REMOTE TRANSMITTER INSIDE VEHICLE.)
t3: APPROX. 30 s
t4: APPROX. 15 s

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*1 : Illumination time can be changed using the Mazda Modular Diagnostic System (M-MDS) or center display (with center display). (See LIGHTING SYSTEM PERSONALIZATION.)